Dilip Rana

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About me

I am a PhD student, researching in the field of power electronics and gallium nitride (GaN) devices. My current research focuses on the reliability characterization of GaN power devices, with a particular emphasis on their operation down to cryogenic temperatures (around 50 K). With my ongoing PhD journey under the supervision of Prof. Zheyu Zhang, I have learned to approach research by tackling technical challenges to design experiments, analyze results and develop scientific judgement. I look forward to further deepening my knowledge in the field of power electronics design, and qualitative and quantitative reliability analysis.

Education

PhD in Electrical and Electronics Engineering (2024 – Expected graduation in 2029)

Rensselaer Polytechnic Institute, School of Engineering, Troy, New York
Term Average (until now): 3.88

Bachelor's Degree in Electrical Engineering (2017 - 2022)

Tribhuvan University, Institute of Engineering, Pulchowk Campus
Term Average: 74.99%

Higher Secondary Education, +2 Science (2015 - 2017)

Premier Higher Secondary School
Term Average: 73.29%

Experience

Graduate Research Assistant: Rensselaer Polytechnic Institute (Jan 2024 – Present)

- ➔ Involved in NASA funded project for performance evaluation of commercial GaN HEMTs
- ➔ Static characterization – extracting IV, transfer and leakage characteristics of commercial GaN HEMTs from major vendors at temperatures ranging from 50-400K
- ➔ Reliability characterization - gate reliability, application replication reliability and power cycling tests of selected GaN candidates
- ➔ Modeling of GaN HEMTs including statistical, dynamic and aging degradation behavior
- ➔ Supported all stages of lab operation and research by collaborating with postdoctoral and fellow lab members

Embedded Hardware Engineer: Yatri Motorcycles (Dec 2022 – Dec 2023)

- ➔ Prototyped and designed of different parts of the vehicle embedded system
- ➔ Performed signal integrity analysis of different boards designed and optimized the circuits
- ➔ Power management system designed with different filters with improved noise immunity and greater system reliability
- ➔ Designed, compared and tested of different methods of user input sensing in industry standard testing conditions
- ➔ Designed and implemented the various parts of the vehicle charging embedded system
- ➔ Supported all stages of embedded system development and collaborated with electrical and mechanical teams of the company

Embedded Hardware Engineer Intern: Yatri Motorcycles (Sep 2022 - Nov 2022)

- ➔ Completed the internship task within specified time and appreciated result
- ➔ Analyzed and designed noise shield for wiring of the vehicle

- ➔ Measured and verified different parasitic parameters of vehicle auxiliaries
- ➔ Designed and tested prototype of the noise filter for different noisy auxiliaries

Publications

- ➔ D. Rana, T. Qiu and Z. Zhang, "A Universal Reliability Platform for GaN HEMTs with Realistic Stress Emulation, Online Monitoring, and In-Circuit Characterization," 2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE), Philadelphia, PA, USA, 2025 open↗
- ➔ A. Siraj, D. Rana, P. Khadka, A. K. Das and Z. Zhang, "Reliability Database for Wide Bandgap Power Semiconductors and Module Packaging via Comprehensive Survey and Standardization," 2025 IEEE Industry Applications Society Annual Meeting (IAS), Taipei, Taiwan, 2025 open↗
- ➔ K. Abinash Man, S. Anjana, R. Dilip and R. Ganesh, "INDUCTIVE POWER TRANSFER: ELECTRIC SCOOTER BASED DESIGN", Final Year Project Report, Apr 2022 open↗

Projects

Static Characterization of Commercial GaN Power Semiconductors (Jan 2024 - Jan 2025)

- ➔ Performed characterization of 14 commercial GaN HEMTs from major vendors including radiation hardened GaN parts for future space application
- ➔ Used cryocooler based setup to achieve temperature from deep cryogenic (50K) to elevated temperature (400K)
- ➔ Characterization of multiple devices considering statistical significance of obtained data, including concurrent characterization of 4 samples at once
- ➔ Extraction of temperature dependency of device performance parameters including – IV, transfer and leakage characteristics

Inductive Power Transfer: Electric Scooter based Design (Jun 2021 - Apr 2022)

- ➔ Performed 2D & 3D electromagnetic simulations and calculated the parameters of various coil designs using COMSOL Multiphysics
- ➔ Manually designed coil with stranded wires and matched up simulated parameters
- ➔ Half bridge inverter with gate driving transformer
- ➔ Power transfer up to 150W within the distance of 20cm and achieved maximum efficiency of 90%
- ➔ Presented the final report to the department with hardware model

Induction Heater Model Design (Nov 2021 - Dec 2021)

- ➔ Prepared the hardware model compatible with market standard coils
- ➔ Presented the hardware prototype at 'Energy Hackathon 4.0'
- ➔ Achieved ZVS by resonance oscillation of the LC tank
- ➔ Power transfer up to 36W with efficiency of 70% compared to resistor heater

Inductive Wireless Phone Charger (Feb 2019 - Mar 2019)

- ➔ PCB embedded pancake coil designed for both transmitter and receiver
- ➔ Colpitts oscillator designed with class-A amplifier to feed the coil
- ➔ Maximum power transfer achieved up to 5W

GSM based GPS Tracker for Electric Vehicle (Jan 2021 - Mar 2021)

- ➔ GPS module and SIM9000 GSM module used with Arduino
- ➔ Auto SMS reading and replying feature with encoded request protection

LCR Meter (Oct 2020 - Nov 2021)

- ➔ L, C & R measured in different ports for each type of components
- ➔ LCD display interfaced with AVR to calculate and display the component values

Spherical Hybrid Black Body Solar (Feb 2018 - Mar 2018)

- ➔ Utilized parabolic reflector to concentrate the light rays to array of spherically arranged 12V solar panels

- ➔ Solar insolation increment up to 50% was achieved

Participations and Awards

Efficiency Theme Hackathon Winner: Energy Hackathon 4.0 - LOCUS (Dec 2021)

- ➔ National level hackathon organized by LOCUS
- ➔ Detailed analysis of 'Transition from LPG to Induction Cooker in Nepal' was done and presented
- ➔ About 30 teams participated within the total of five different themes

Participations: LOCUS Exhibition 2018, 2019 & 2020 - LOCUS

- ➔ National level project exhibition and competition organized by LOCUS every year
- ➔ Presented the hardware model of 'Spherical Hybrid Black Body Solar' at LOCUS 2018
- ➔ Presented 'Inductive Wireless Phone Charger' at LOCUS 2019
- ➔ Designed and Presented 'LCR Meter' at LOCUS 2021
- ➔ More than 100 teams participated every year

Participation: e-Yantra Robotics Competition (eYRC 2019-20)

- ➔ International level robot building competition organized by IIT bombay
- ➔ Developed an automatic object picking and transporting robot within predefined routes

Honorable Mention: NASA Space Apps Challenge - NASA (Oct 2020)

- ➔ International level online apps challenge
- ➔ Developed 'Solar Sailer', space exploration simulation tool with simulated solar system and effects like time dilation, doppler effect and view distortion

Overall and Health Theme Hackathon Winner: Quantum Hackathon - NxtGen (Sep 2020)

- ➔ National level online hackathon organized by NxtGen
- ➔ Worked on and presented 'Masked', a python-based face mask detecting application

Skills

1. Hardware Design: Circuit design, assembly & troubleshooting
2. Equipment/Measurement: Curve tracer, Oscilloscope, Helium based cryocooler
3. Simulation Tools: MATLAB simulink, PLECS, LTSpice, Saber, Proteus, Multisim, Circuit Wizard
4. Circuit Fabrication: Altium Designer, KiCAD, EasyEDA, Circuit Wizard
5. FEA Tools: COMSOL
6. CAD Tools: SOLIDWORKS, AutoCAD
7. Microcontrollers: TI DSP, STM32, AVR, Arduino
8. Programming: MATLAB, MATLAB GUI, C, C++, Java, Embedded C

Areas of Interest

- ➔ Wide Bandgap Power Semiconductors (GaN, SiC)
- ➔ Reliability Assessment of Power Semiconductors and Converters
- ➔ Automated Test Setups, Design of Experiments
- ➔ Cryogenic Power Applications
- ➔ Converters Design, Motor Drives
- ➔ Energy Storage and Transfer System

Involvements

- ➔ APEC 2026 (March 2026): Student Session Assistant
- ➔ MATLAB Simulink Workshop (Nov 2021): Lecturer
- ➔ Basic Electronics and Proteus Workshop (Jan 2022): Lecturer
- ➔ Electrical Club of Pulchowk Campus (2020): Event Manager